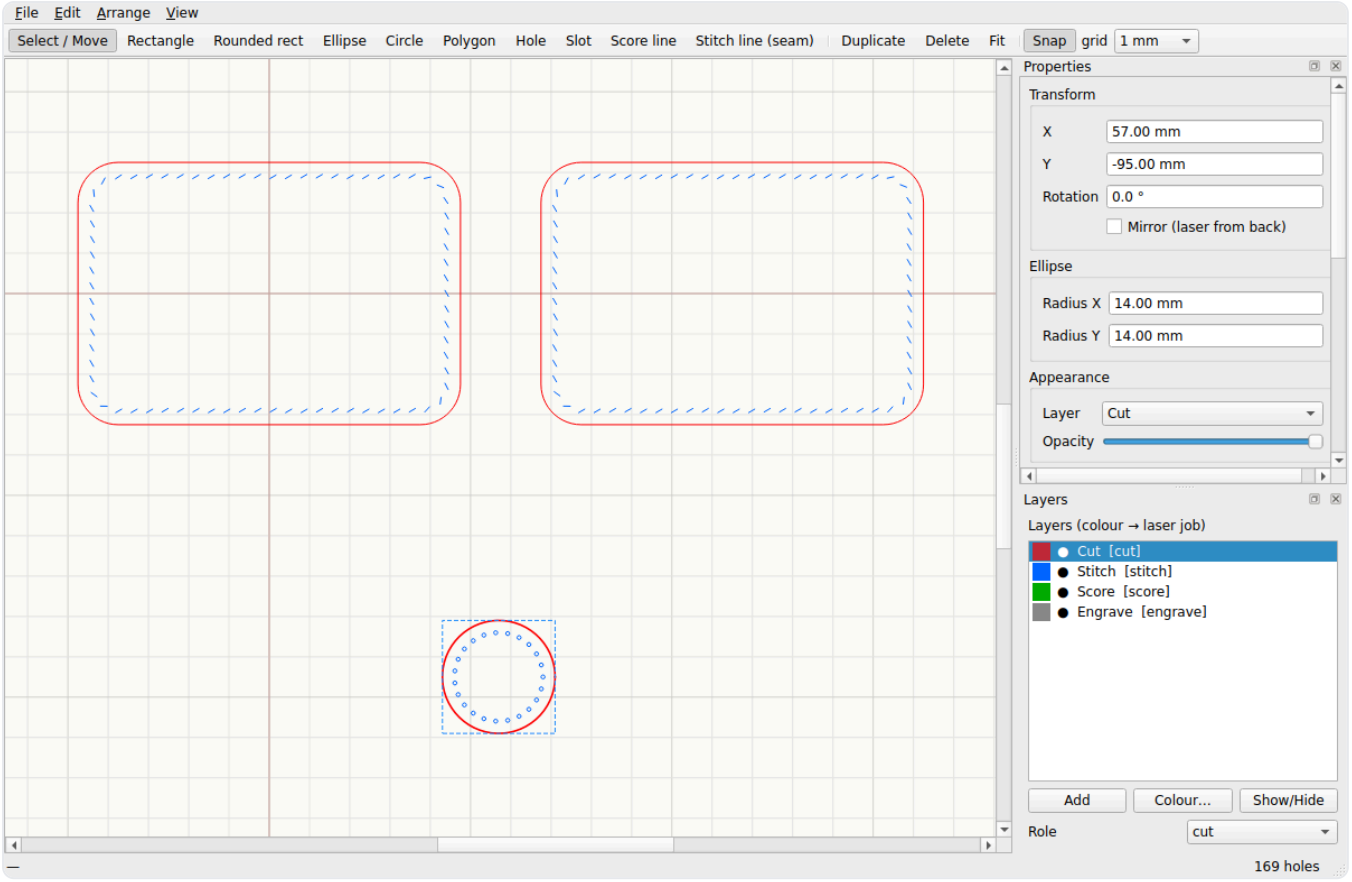
Stitch Hero

A Python desktop **CAD program for laser-cut leather patterns**, built around the one thing most pattern software gets wrong for hand-stitchers: **stitch hole spacing that matches a real pricking iron**, and holes that **line up perfectly across overlapping pieces** no matter which side you laser.



> Status: v0.3, working and tested. Draw with a full CAD toolset (rectangles, circles, polygons, bezier pen, arcs, trim, fillet / chamfer, boolean ops), get chord-accurate stitch holes that register hole-for-hole across pieces, and put laser-ready **SVG / DXF / 1:1 PDF** on the bed — with kerf compensation, sheet nesting, a parts library, starter templates, job & thread estimates, autosave and a dark theme. Roadmap below.

📖 **New here? Read the User Guide** — a full manual with tool reference, a stitching deep-dive, three worked tutorials (card holder, curved key fob, shared seams), shortcuts, and troubleshooting. It's also built into the app: **Help → User guide** or press **F1**. The test suite pins the manual to the code, so the docs can't silently drift out of date.

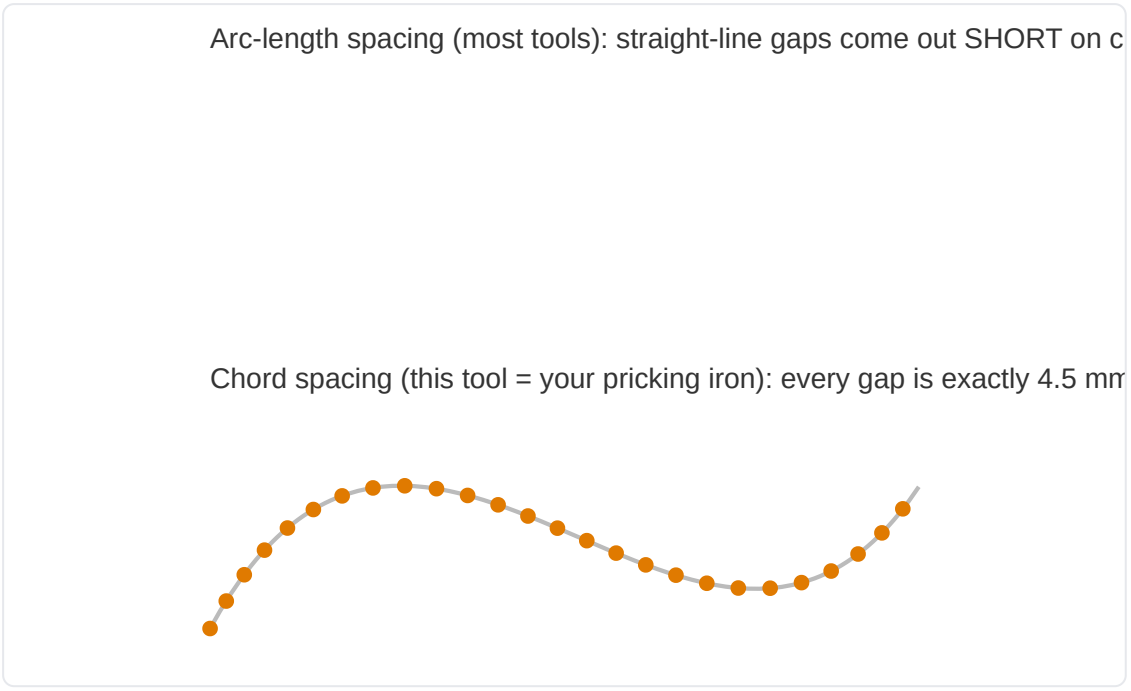
📄 **Prefer a printable copy? Download the manual as a PDF.**

Two problems this is designed around

1. Pricking-iron spacing, not contour spacing

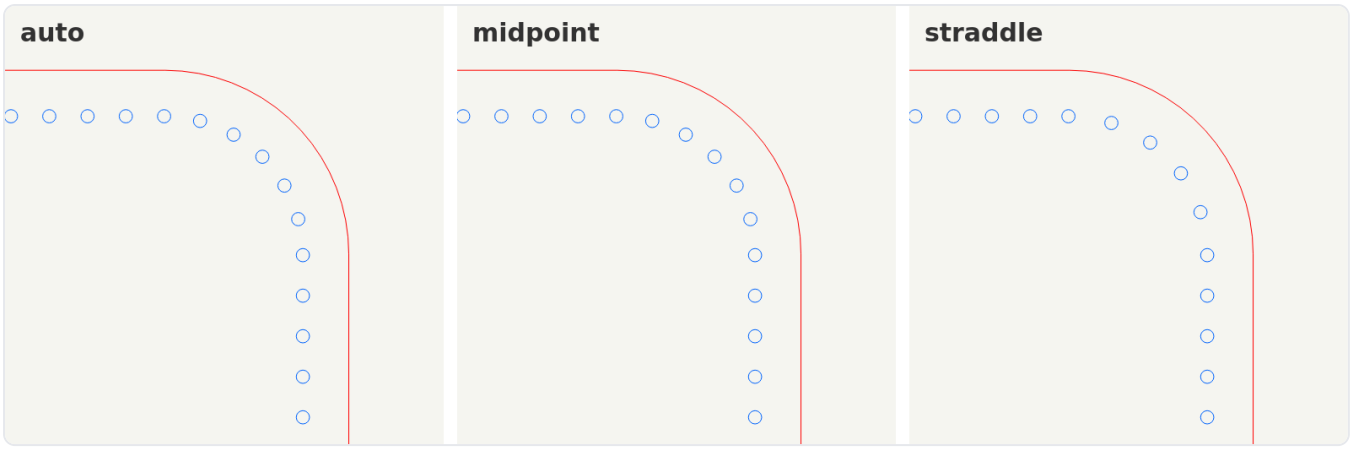
Ask most tools for "3.85 mm spacing" and they drop a hole every 3.85 mm *of arc length* along the contour. On a curve the actual straight-line gap between neighbouring holes then comes out **short**, so the holes never match your physical iron, whose teeth are rigid and a fixed **straight-line (chord)** distance apart.

Stitch Hero does **chord marching**: from each hole it finds the next point forward along the path whose straight-line distance is exactly the iron's pitch. Consecutive holes are always `pitch` apart point-to-point — exactly what the iron does as you rotate it around a curve.



It also nudges the pitch to the nearest whole-hole count so holes land cleanly on **every corner** and on **both ends** of an open seam — the thing leatherworkers do by hand — and the readout tells you how far it had to move.

On a **rounded corner** the holes come out **symmetric about the arc's midpoint** — either a stitch sitting right on the 45° apex or an even pair straddling it, never a lone hole landing off-centre. A generous corner is fitted as its own span; a **tight corner** collapses to a single clean hole on the apex instead of cramming several holes into a couple of millimetres. Pick `auto` (whichever count best matches your iron), `midpoint`, or `straddle` per shape. A happy side effect: a symmetric outline now gets flip-symmetric holes for free.



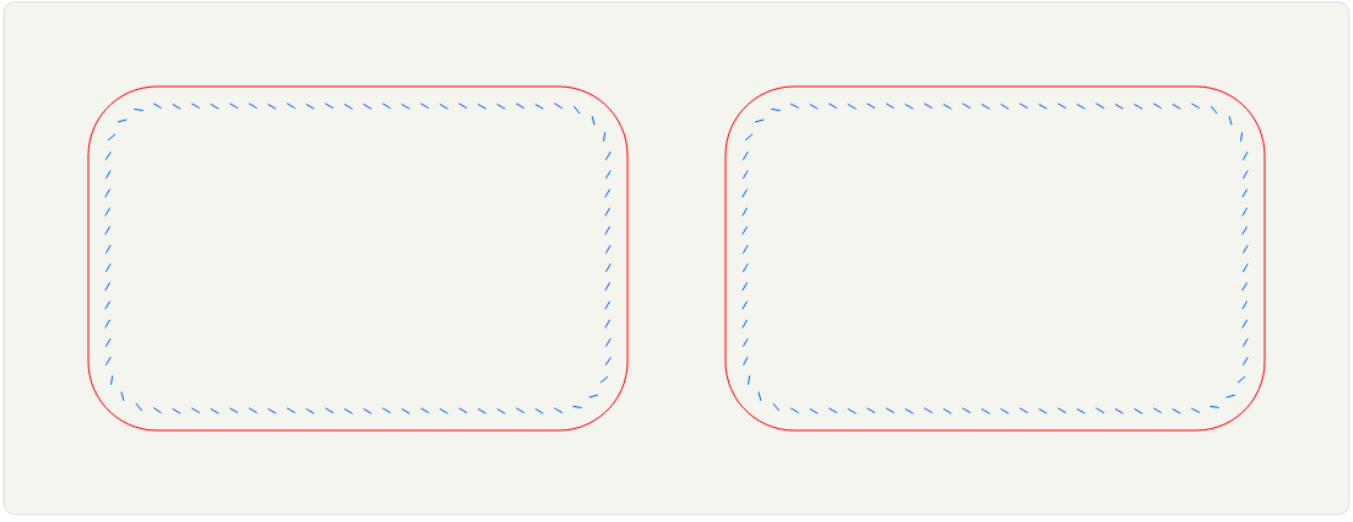
2. Registration across overlapping pieces

Two pieces stitched together (front + lining, gusset + panel) must have their holes in **identical positions** or they won't line up. Stitch Hero guarantees this three ways:

- **Deterministic perimeter stitching** — two pieces with the same outline and settings get *byte-identical* hole layouts automatically. Duplicate a panel and the holes already match.
- **Mirror-safe** — flip a piece to laser it from the back (`Mirror`) and the holes stay registered, hole-for-hole.
- **Shared stitch line (seam)** — for pieces with *different* outlines that share one edge, draw one seam and every piece against it gets the same holes.

Verify any of it visually by dragging one piece on top of another and dropping the opacity — the tool is built for exactly that overlay check.

Need the matching reverse-side piece? **Make back piece** (`Ctrl+M`) drops a mirror-image copy whose holes stay registered hole-for-hole — ready to laser from the back. The slanted slits mirror too, so they line up when the piece is flipped:



Install & run

Download a ready-to-run build

From the **Releases** page — no Python needed:

- **Windows** — `Stitch-Hero-Setup-<version>.exe` (per-user install, no admin prompt).
- **macOS** — `Stitch-Hero-<version>-macOS-universal.dmg` (one download; runs natively on both Apple Silicon and Intel). Open it and drag the app to Applications. It's unsigned, so the first launch is **right-click → Open** (or `xattr -dr com.apple.quarantine "/Applications/Stitch Hero.app"`). *Mac builds are paused for now: releases after 0.3.2 ship Windows only unless the build is run with "Also build the macOS .dmg" ticked.*

Both are built by the **Release builds** GitHub Action: push a `v0.2.0`-style tag, or run it manually (Actions → *Release builds* → *Run workflow*), and it publishes the Windows `.exe` (and the macOS `.dmg`, when that option is ticked) to a Release automatically. What changed in each version is in CHANGELOG.md.

From source (any OS)

```
git clone https://github.com/heathmanc/leather-drafting.git
cd leather-drafting
pip install -e ".[gui]"        # installs PySide6; Python 3.9+
python -m leathercad_app       # or: stitch-hero
```

On **Windows/macOS** the PySide6 wheel is self-contained. On headless **Linux** you may need system libs: `sudo apt-get install libegl1 libgl1 libxkbcommon0`.

The engine alone (spacing, geometry, export) has **zero dependencies** — you only need PySide6 for the GUI.

Using the app

The drawing tools live in a **vertical tool palette docked on the left**. It is a normal draggable toolbar — grab its handle to move or float it, drop it on any edge, and use the **pin** button at the top to lock it in place. Its position (and the window layout) is remembered between sessions.

Tool palette	Related shape tools collapse into **fan-out buttons** to keep the palette tidy — **Rectangles** (rectangle / rounded), **Circles & ellipse** (circle / ellipse / 2-point / 3-point) and **Arcs** (3-point / centre). Click the button for its current variant, or the little ▸ arrow to pick another; the button then remembers your choice. Every variant still has its own keyboard shortcut
Draw	Rectangle `R`, Rounded rect `O`, Ellipse `E`, Circle `C`, Polygon `P` (click points, double-click to finish), Line `L` (2-point)
Circle by points	**2-point circle** `2` (click the two ends of a diameter) and **3-point circle** `3` (click three points on the rim) — snap them onto existing geometry to fit a circle exactly
Arc	**3-point arc** `4` (click the start, a point on the arc, then the end) and **centre arc** `5` (click centre, start, then end — sweeps counter-clockwise). Arcs are kept as real arcs (an editable `EditablePath`), so double-click to node-edit and drag the midpoint handle to reshape; they stitch and export like any edge
Pen / bezier	Pen `B` — click to drop a **corner** anchor, or **click-drag** to pull out a smooth tangent handle (standard vector-pen behaviour; drag farther for a deeper curve). Consecutive anchors are joined by **cubic bezier curves**. **Right-click** (or `Enter`) finishes an open curve; **click the first anchor** to close it; `Esc` cancels. The result is a node-editable curve: enter node-edit (double-click) and drag the green **control handles** to reshape each curve, or drag an anchor and its handles follow. Curves get default stitching like any shape (toggle it off in Properties), and flatten cleanly for chord-accurate holes and export
Construction line	`G` — a dashed guide you drag out; it's a snap/alignment reference only, never cut or exported
Drawing style	**every drawing tool places points by clicking** — no click-and-hold. Two-point tools (rect, rounded, ellipse, circle, slot, line, construction) take two clicks; polygon / seam / score take clicks then a double-click (or `Enter`); hole is one click. Prefer press-drag-release for the two-point tools? Toggle **View → Drag to draw** (remembered between sessions). Hold **Shift** while placing any line — line, construction, score, or seam — to lock the segment to **0° / 45° / 90°** from the previous point. `Esc` cancels a point you're mid-placing; press `Esc` again (nothing in progress) to drop back to the pointer
Holes / slots / fold lines	Hole `H` (hardware), Slot `T` (stadium), Score line `K` (fold/skive on the Score layer)
Seam	Stitch line `M` — a shared seam for cross-piece registration
Select / move	`S` — drag to move, drag one piece over another to check fit
Line width	toolbar **line** spinbox sets the on-screen stroke width for outlines, seams, dimensions and text (remembered between sessions)
Snapping	two independent toolbar toggles — **Nodes** and **Grid** — plus grid size. Node snap catches real geometry: **endpoints, edge & arc midpoints, arc/shape centres, circle quadrants, stitch-hole centres, intersections, and the midpoints of the pieces an intersection carves out** (bisect a line and you can grab its quarter points) — no phantom bounding-box points. Turn Grid off to snap only to geometry (points off a node stay free). Both apply while drawing and moving. Dragging a **circle** locks by its **centre** — the centre wins over the quadrants so you can drop it straight onto another node. **Construction/guide lines** also snap **anywhere along their body** (nearest-point-on-line), and where a guide crosses a line the **junction** splits it so the resulting sub-segment midpoints are snappable from anywhere along them

Alignment guides	while drawing, when the cursor lines up with another object's node/centre the point locks to that x/y and a dashed **guide line** appears (smart snapping, Fusion/Illustrator style)
Exact sizes	live W×H shown while dragging; after drawing, the size field is focused so you can type an exact value
Resize handles	select a single rectangle / circle / ellipse and drag one of the **8 box handles** (4 corners + 4 edge midpoints) to resize by eye — the opposite corner/edge stays pinned. Or type exact dimensions in Properties; both stay in sync
Select by outline	shapes are grabbed by clicking their outline, not the filled interior — click "inside" to reach shapes behind or draw there
Edit nodes	double-click a polygon/seam, or right-click a shape → *Convert to editable nodes* (`Ctrl+K`). Rounded corners keep their **arcs**: each arc shows its two endpoints plus a midpoint handle you drag to reshape the curve. The shape locks while editing so clicks grab the nodes. Hold **Shift** while dragging a node to lock its movement to **0° / 45° / 90°** from where the drag began (so an endpoint drags straight up into a vertical line).
Line length / angle	select a line and type its **Length** and **Angle** in Properties (the first point stays put) — not just its x/y position
Snap to nodes	with Snap on, dragging a shape (or a node while editing) magnetically snaps to other shapes'/holes' nodes
Break apart	right-click a shape → *Break apart into segments* (`Ctrl+B`) — each edge (line or arc) becomes its own movable piece
Join / weld	select 2+ pieces → *Join / weld segments* (`Ctrl+J`) — chains segments whose endpoints touch into one path (arcs kept); loose holes in the selection attach to the piece they're inside
Trim	Trim tool (`X`) — click the part of an outline you want gone; it's cut back to wherever it crosses another shape, just like Fusion 360 / LightBurn. Arcs are preserved; a closed shape opens, an open one splits
Text	Text tool (`A`) — click to place **engrave lettering**; glyph outlines are baked to paths on the Engrave layer and export as engrave polylines
Measure	Measure tool (`Q`) — click two (snapped) points for a live **length / angle / dx / dy** readout in the status bar
Dimension	Dimension tool (`D`) — click two points to drop a permanent **dimension annotation** (extension lines, arrows, measured length). Snap the ends onto a shape and the dimension **tracks it**: move or resize the shape and the length updates. Annotations only — never cut or exported
Group (move together)	select two or more items (holes, shapes, seams) → **Group** (`Ctrl+G`) — they share a group and move as one unit; clicking any member selects the whole group. **Ungroup** (`Ctrl+Shift+G`) breaks it apart. Survives save/undo
Attach holes to shape	*Attach holes to shape* (`Ctrl+Shift+A`) bakes loose holes into a shape so they ride with it for cross-piece registration — select several pieces (or everything) and each hole goes to the piece it's inside; *Explode stitching → holes* turns a shape's stitching back into individual, deletable holes (they stay in the piece's move-group)
Right-click menu	Group / Ungroup / Attach holes / Convert to nodes / Duplicate / Delete on the selection

Radius corners	select a rectangle/polygon, set *Corner radius* in Properties
Iron & holes	per shape: pick an iron from the standard mm ladder (SPI shown) or type any pitch (`25.4/7` works), inset, round or slanted-slit holes, single or **double row** (saddle stitch) + backstitch on open runs, live hole count + spacing readout
Corner holes	rounded-corner holes are always **symmetric about the arc midpoint**. Stitching → *Corners*: `auto` (best count for your iron), `midpoint` (a hole on the apex), `straddle` (an even pair around it)
Back-to-back symmetry	Stitching → *Symmetry* (vertical/horizontal) forces flip-symmetric holes; Edit → *Check back-to-back symmetry* validates that a flipped piece lines up
Make back piece	Edit → *Make back piece* (`Ctrl+M`), or right-click → *Make back piece (mirror)* — drops a mirror-image copy whose holes stay registered with the front so the two stitch together back-to-back
Layers → laser jobs	colour-coded Cut / Score / Engrave / Stitch layers with **Show/Hide** visibility. A piece's blue stitch holes follow the **Stitch** layer, so you can hide **Cut** to see just the stitch pattern (or hide Stitch to see the bare outline)
Arrange	align (left/centre/right/top/middle/bottom) and distribute selected shapes
Undo / redo	`Ctrl+Z` / `Ctrl+Shift+Z`
Duplicate / Delete / Fit	`Ctrl+D` / `Del` or `Backspace` / `F`
Zoom / pan	mouse wheel / middle-drag
Save / Open	`Ctrl+S` / `Ctrl+O` (JSON project files)
Export	SVG `Ctrl+E` or DXF — millimetre-accurate, layer-coloured. SVG layers are named and ordered **holes** first, **outline** last; the holes are LightBurn's own blue, so in LightBurn they land on layer C01, ahead of the red outline on C02, and are cut first. DXF outlines export as connected **POLYLINE**s (closed shapes carry the closed flag), so LightBurn / Illustrator import them as one contour instead of loose line segments
Print / PDF (1:1)	**Export PDF (1:1, tiled)** and **Print** `Ctrl+P` render the pattern at **true 1:1 scale** (1 mm = 1 mm on paper). Patterns bigger than a sheet are split into overlapping **tiles** with crop marks and `row/col` page labels — line the pages up on the overlap and tape. Each page prints a labelled reference box so you can confirm your printer isn't scaling

Everything is in **millimetres**, Y-up, and the canvas is WYSIWYG with the export.

Trim

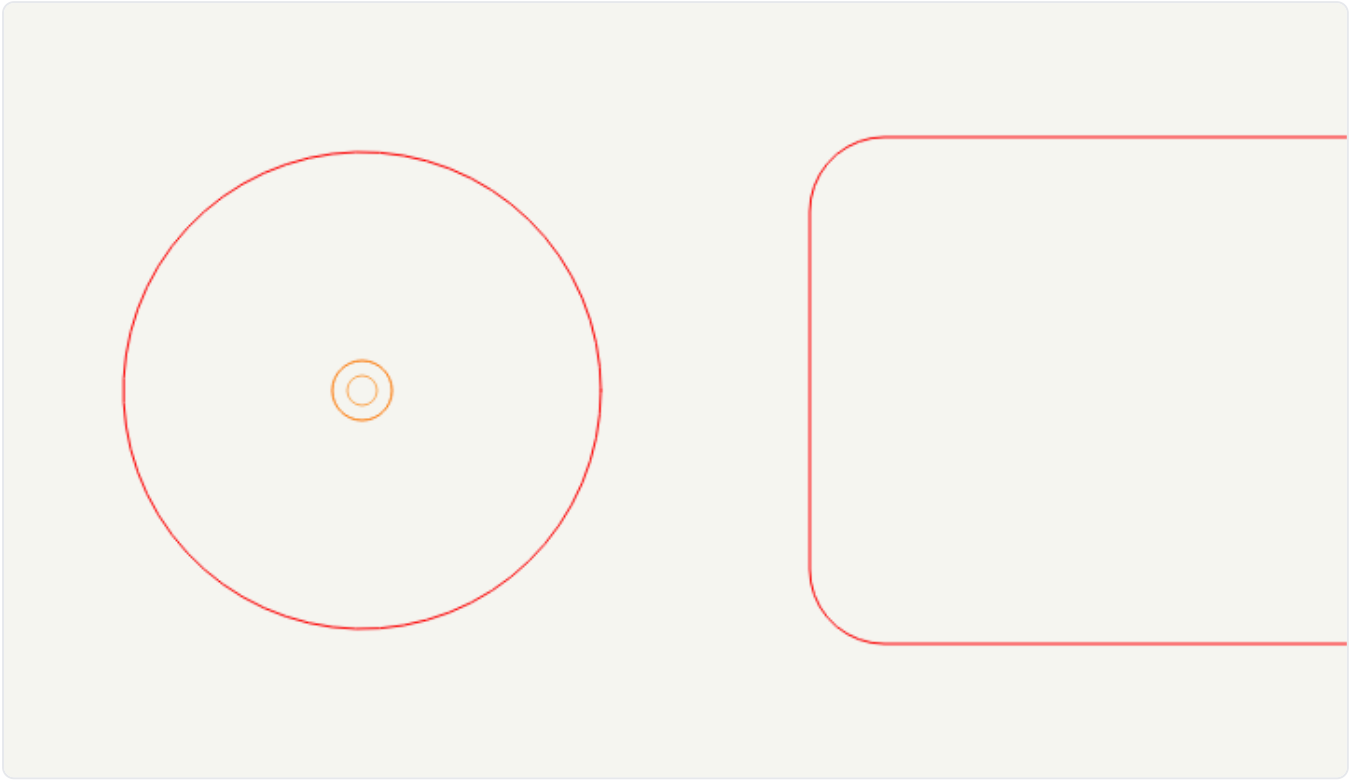
Pick the **Trim** tool (`X`) and click the part of an outline you want gone. It is cut back to the two points where it crosses other geometry — the classic 2D-sketch trim from Fusion 360 / LightBurn. Curve type is kept (a trimmed arc stays an arc); a closed shape opens up, and an open path clicked in the middle splits into two. Clicking a piece that nothing crosses removes it outright. Cuts land on the true lines, arcs and curves, and a line drawn to end on the outline shares its end point with the trimmed piece, so the pieces join and fillet cleanly.



Smart snapping & construction lines

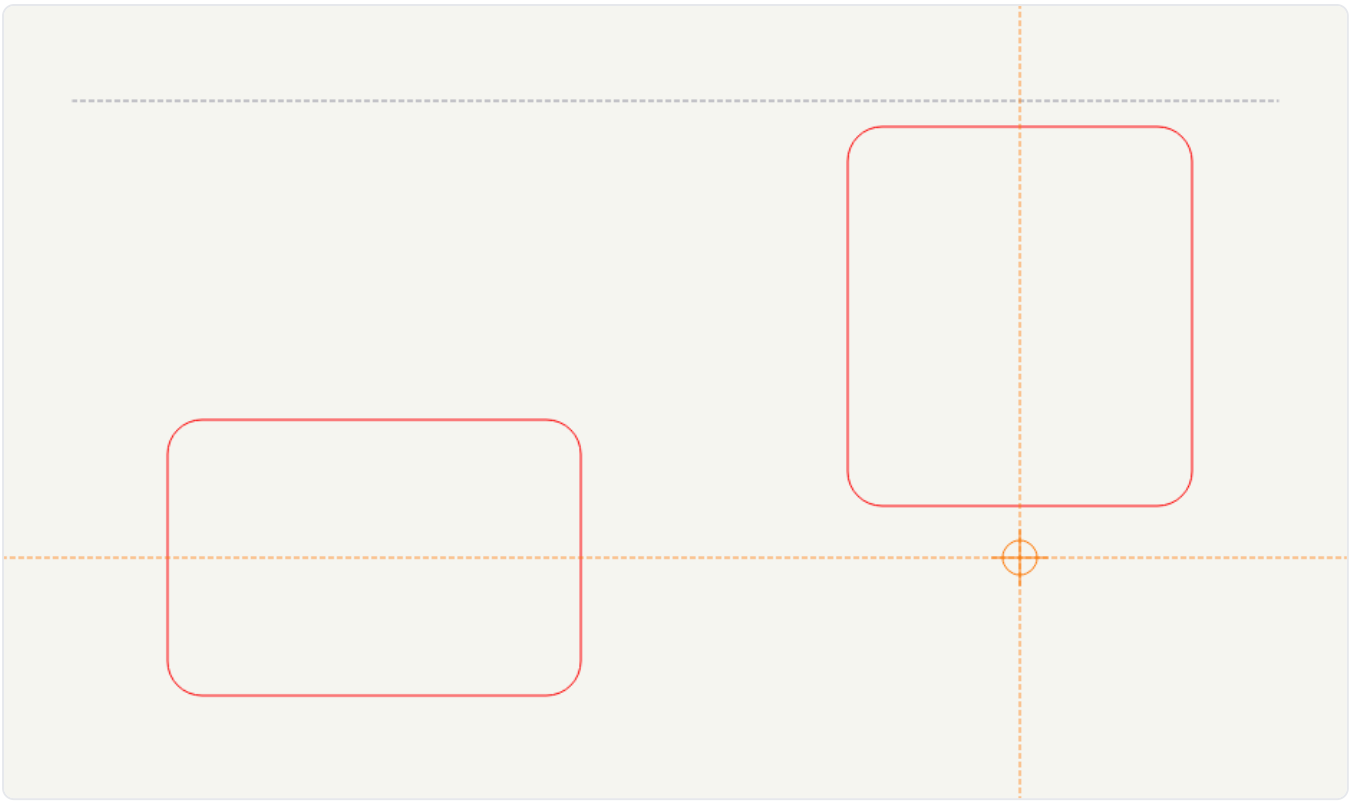
While a drawing tool is active, the snap targets near the cursor are **marked on screen** so you always know where a click will land — a **square** for an end/corner, a **diamond** for a midpoint or circle quadrant, a **circle** for a shape/arc centre, a small **ring** for a stitch-hole centre, an **×** for an intersection — and the point you're placing snaps to the nearest one (its type is named in the status bar). All targets come from real geometry, so nothing floats off a rounded or round outline.

When the point lines up (same x or y) with a nearby **endpoint, midpoint or centre**, it **locks to that alignment** and a dashed orange **guide line** is drawn — the same inference snapping as Fusion 360 / Illustrator. (Stitch holes and intersections don't drive guides, so the lines stay meaningful even on a piece with a hundred holes.) Line up with two things at once and it snaps to the crossing point.



Node snapping and grid snapping are **separate toolbar toggles** (*Nodes* and *Grid*). Turn *Grid* off to snap purely to geometry — points that aren't on a node then stay wherever you put them.

Drag out a **construction line** (`G`) for a reusable guide of your own: it's dashed, snappable, and never cut or exported.

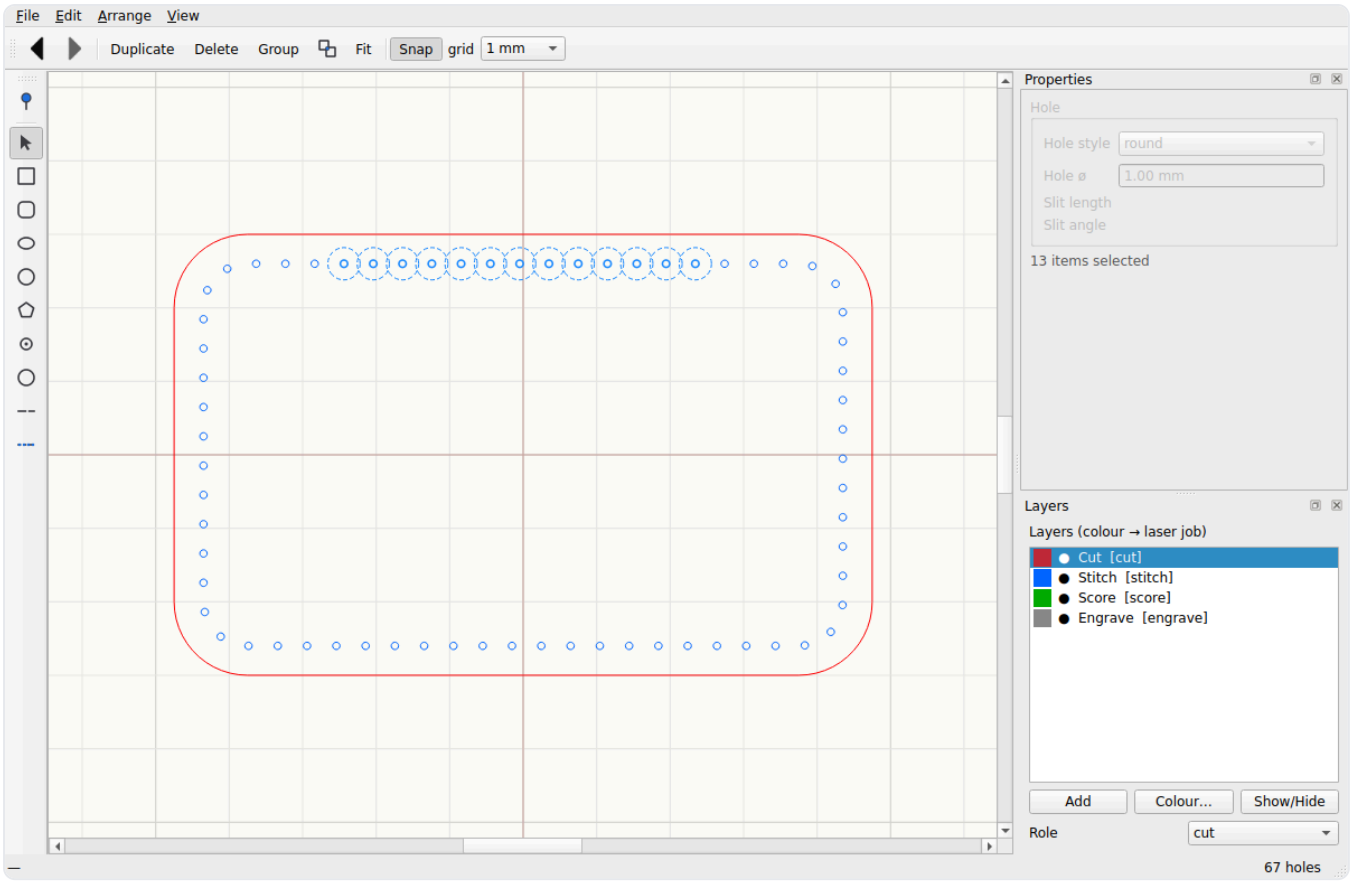


Group / ungroup holes (removing individual ones)

By default a shape's holes are *computed* from its stitch settings, so they stay evenly spaced when you change the iron or resize — but that means the engine would redistribute them if you deleted one. To hand-edit holes (say, to clear a spot for a rivet):

- **Ungroup** (right-click → *Ungroup stitching*, or `Ctrl+Shift+G`) explodes the shape's holes into **individual holes** and turns its auto-spacing off. Each hole is now an ordinary object — **click one to select it** (rubber-band to select many), drag to nudge, press **Delete** to remove. Nothing reflows; the gap stays exactly where you made it.
- **Group** (right-click → *Group holes into shape*, or `Ctrl+G`) does the reverse: select a shape **and** the loose holes, and they get baked back into the shape so they move and rotate with it as one unit — still without redistribution.

Right-click anywhere for the context menu (Group / Ungroup / Duplicate / Delete).



More in the app

Beyond drawing and stitching, Stitch Hero covers the rest of a real bench workflow:

Fillet / chamfer corners	Per corner, a **round** (fillet arc) or a **bevel** (straight chamfer) — set the radius or setback in the toolbar box, then click corners
Boolean ops	**Union**, **difference** and **intersect** on closed shapes — cut windows and card slots straight out of a panel
Offset / seam allowance	Grow or shrink an outline by a fixed amount for a true seam allowance
Import SVG / DXF	Bring existing vector drawings in as fully editable shapes — paths, arcs and colours map onto layers
Tracing image	*View → Tracing image*: drop a photo or scan behind the canvas, **calibrate it to real millimetres** (two clicks + a known distance), fade it, and trace over it with the normal tools
Nest on sheet	Shape-aware nesting packs your pieces onto a leather sheet; each piece's slots, holes and seams travel with it
Kerf compensation	On export, closed **cut** outlines are compensated for the beam — outer edges grow by kerf/2 and inner cut-outs shrink — so parts come out drawn-size
Job estimate	Piece and hole counts, thread, cut / score / engrave travel, leather area used (with waste), and — when you feed it prices — a leather / thread **cost** breakdown
Thread estimate	Saddle-stitch thread length for a piece, from its holes, stack thickness and tail allowance
Seam-mate check	Verify that pieces sharing a seam actually line up hole-for-hole before you cut
Parts library	Save any selection to a personal parts library and drop it into any document
Templates & generators	Open a ready-made card holder, belt, key fob or slim card holder, or generate parametric **card-pocket stacks** and **zipper openings**
Stitch only some edges	right-click an edge of a stitched piece → *Don't stitch this edge*: the seam stops with a hole at the inset corner either side of a pocket's open mouth
Hardware library	rivets, snaps, Chicago screws, Sam Browne studs, magnetic snaps, eyelets, buckle slots, turn locks and ring footprints in the Parts dock — each placed with the cut it needs and a **stitch keep-out** the seam flows round
Group rotate / scale / flip	one rotate grip for a group or multi-selection; Arrange → Rotate 90°/180°/by..., Scale..., Move by..., Flip horizontal / vertical, Mirror across selected line; arrow-key nudge
Units	View → Units → Inches shows and takes every length in inches (stored in mm); type `10mm` to enter millimetres
Fold allowance	Edit → Fold allowance... lengthens a piece across its fold lines by angle × (radius + ½ thickness)
Generators	strap / belt, box and curved gussets, bifold wallet, key tab, hemmed pocket — plus card pocket stacks and zipper openings
Export options	outlines / holes / lettering, selection only; arcs and curves export as true arcs (SVG `A` / `C`, DXF bulges)
Edit several at once	select several pieces; only the fields you change are written to all of them · Layers panel **Lock**

Dark theme	A full dark UI for late-night bench sessions (*View → Dark theme*)
Autosave & recovery	Work is autosaved and offered back after an unexpected quit

Scripting API (no GUI needed)

The whole model is usable headless — handy for parametric patterns:

```
from leathercad import Document, Rectangle, Transform, StitchSettings, get_iron, export

doc = Document("wallet")
iron = get_iron("3.85mm")
for x in (0, 120):                      # front + lining, placed apart
    doc.add_shape(Rectangle(
        width=95, height=65, corner_radius=10,
        transform=Transform(x=x, y=0),
        stitch=StitchSettings(pitch_mm=iron.pitch_mm, inset=3.5,
                              hole_style="slit", slit_angle=30),
        layer="Cut"))

export.export_svg(doc, "wallet.svg")
export.export_dxf(doc, "wallet.dxf")
doc.save("wallet.leathercad.json")
```

Runnable examples:

```
python examples/wallet_document.py   # builds a doc, proves registration, exports
python examples/card_holder.py       # engine demo: chord vs arc-length on a curve
```

Tests

```
pip install -e ".[dev]" && QT_QPA_PLATFORM=offscreen pytest
```

The suite proves the guarantees that matter: chord spacing equals the iron pitch on curves while arc-length spacing is measurably short; identical and mirrored pieces get registered holes; save/load and SVG/DXF export round-trip.

Project layout

```
leathercad/            pure-Python model + engine (no dependencies)
  geometry.py          Vec2 / vector math
  path.py              segments (line/arc/bezier), Path, PathBuilder, flattening
  offset.py            inward polygon offset (stitch-line inset)
  stitching.py         chord/arc marching + fitting + registration  <- the core
  stitchsettings.py    per-shape/seam stitch config
  shapes.py            Rectangle/Ellipse/Circle/Polygon/PathShape/EditablePath (line/arc/bezier)
  layers.py            colour layers -> laser jobs
  stitchline.py        shared seam for cross-piece registration
  document.py          the CAD document + JSON save/load
  irons.py             pricking-iron pitch / SPI presets
  export.py            SVG + DXF exporters (with kerf compensation)
  trim.py              trim an edge back to its intersections
  modify.py            per-corner fillet (round) / chamfer (bevel)
  boolean.py           union / difference / intersection on polygons
  importers.py         import SVG & DXF as editable shapes
  nesting.py           pack pattern pieces onto a sheet
  estimate.py          leather-area / travel-length / cost job estimate
  thread.py            saddle-stitch thread-length estimate
  templates.py         starter documents (card holder / belt / key fob …)
  generators.py        parametric part generators (card pockets, zippers)
  text.py              engrave lettering baked to paths
  dimension.py         dimension annotations that track shapes
leathercad_app/        PySide6 desktop app
  canvas.py            mm-accurate Y-up QGraphicsView, tools, grid, zoom/pan, tracing underlay
  items.py             shape/seam graphics items with live hole rendering
  panels.py            properties + layers docks
  partspanel.py        parts-library dock
  theme.py             light / dark themes
  robustness.py        autosave + crash recovery
  mainwindow.py        window, toolbar, menus, export wiring
  __main__.py          python -m leathercad_app
examples/  tests/  docs/
```

Roadmap

- [x] Draw rectangles / rounded rects / ellipses / circles / polygons
- [x] Radius (fillet) corners
- [x] Move & overlay shapes (opacity) to verify sizing
- [x] Per-shape pricking-iron sizes (mm or SPI), round or slanted-slit holes
- [x] Chord (iron-accurate) spacing with corner/endpoint fitting
- [x] Cross-piece registration (deterministic + mirror-safe + shared seams)
- [x] Colour layers → laser jobs

- [x] SVG + DXF export, JSON project save/load
- [x] Undo/redo history
- [x] Grid + vertex snapping, live and type-in dimensions
- [x] Align / distribute
- [x] Edit polygon/seam vertices; add holes / slots / skive (score) lines
- [x] Two-row saddle stitch + backstitch markers
- [x] Left tool palette (draggable / floatable / pinnable, layout remembered)
- [x] Group / ungroup holes (individual selectable holes; group back to a shape)
- [x] Right-click context menu (group / ungroup / convert-to-nodes / duplicate / delete)
- [x] Convert a shape to editable nodes (rounded corners keep arcs: midpoint + endpoint handles); outline-based selection; node-to-node snapping (move + edit)
- [x] Break a shape apart into movable line/arc segments, and join/weld them back
- [x] Flip-symmetric hole distribution + back-to-back symmetry validation
- [x] Symmetric rounded-corner holes (apex / straddle) + make-back-piece (mirror)
- [x] Trim to intersections (Fusion / LightBurn style), arcs preserved
- [x] Alignment guides (smart snapping) + Line and construction-line tools
- [x] Fillet & chamfer (round / bevel) corners
- [x] Boolean ops (union / difference / intersect) + offset / seam allowance
- [x] Import SVG & DXF drawings as editable shapes
- [x] Place, calibrate and trace over a raster reference image
- [x] Kerf compensation on cut outlines
- [x] Nest pattern pieces onto a leather sheet
- [x] Job estimate (leather area + waste, travel lengths, optional cost)
- [x] Thread-length estimate + seam-mate check
- [x] Parts library, starter templates, and parametric generators
- [x] Dark theme, autosave & crash recovery
- [x] Print-to-scale PDF tiling for hand cutting

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